AI-Optimized Inventory Management Systems in Construction

AI Digital Transformation Case Study of an ERP-Integrated, AI-Driven Construction Management Platform

MBA Study Report

Course: AI in Business

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Note: This research examines how artificial intelligence could improve construction inventory management by integrating planning, cost control, procurement, warehouse operations, and field execution into a single digital platform.

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Abstract

1.1 *In the scenario, a big contractor uses corporate systems for budgeting and buying, a scheduling platform for base tasks, and site teams that send daily progress updates. Despite these procedures, delivery is still being delayed by material shortages, too much inventory, delayed approvals, and a lack of insight into the difference between scheduled demand and storage capacity.*

The research asserts that the principal issue is not the absence of data but the inadequacy in integration, governance, and predictive capabilities. A proposed AI-enabled construction management solution has a control matrix that brings together bill of quantities items, work breakdown structure codes, activity IDs, locations, productivity requirements, and material specifications. The program generates work packages based on lookahead, makes it easier to test and approve quality at checkpoints, tracks actual quantities and man-hours, and connects these operational records to a SQL Server data warehouse. The warehouse combines baseline amounts from Primavera P6, budget and actual cost data from SAP, procurement and inventory transactions, inspection findings, and RFID-enabled labor tracking. With this architecture, the company can move from managing its inventory reactively to optimizing it based on predictions. Machine learning models can tell you how much material you need and how likely you are to run out of it. Rule engines can check requests and approvals, robotic process automation can help with tasks that need to be done over and over again, and analytics dashboards can show you how ready your materials are, how much you've committed to them, how much you've actually used them, how much value you've earned, how productive you've been, and how much you expect to spend. The case shows that AI doesn't replace project controls; instead, it improves them by enhancing timing, visibility, and decision quality. The study says that measurement governance, change control, master data discipline, cybersecurity, and phased deployment are all necessary for successful implementation. These results align with the assignment's need to integrate AI, analytics, digital transformation, and ERP into a cohesive business case.

1. Industry and Business Context

2.1 The construction *business is one of the least computerized fields, even though project success depends on careful coordination of materials, labor, equipment, subcontractors, and funds. McKinsey notes that engineering and construction projects often run late and go over budget, and that the industry remains relatively digitally immature. The study also shows that digital transformation can yield significant gains in productivity and cost savings when pursued alongside changes to how the business operates, rather than as separate technology projects (Koeleman et al., 2019; McKinsey, 2019).*

2.2 In fact*, this means contractors often have many systems but still don't have a clear picture of how their operations are going. This is especially important in construction because goods are expensive and time-sensitive. Building supplies, on the other hand, are used in particular orders, linked to specific locations, checked, and changed by changes, weather, logistical problems, and design changes. An enterprise may appear sufficiently equipped with ERP; nonetheless, it may still be incapable of executing a task due to misallocation of materials to the correct work package, quality approval, or distribution to the pertinent region. The difference between accounting visibility and execution readiness is important in this case.*

2.3 The issue *at hand is a contractor that bids on and completes projects using a well-known pre-contract and post-award workflow. During the bidding process, the company receives the client's bill of quantities and uses an estimating platform like Candy to develop resource-based unit pricing. After the award, planners use the Work Breakdown Structure (WBS) in SAP to convert the estimate into control budgets, and they use Primavera P6 to convert the same scope into baseline activities and quantities. After that, site engineers carry out these responsibilities by planning ahead, reporting daily, and requesting materials. Even though each part makes sense on its own, the company still faces material shortages, manual reconciliation, and delayed managerial insights because the system architecture lacks a single place for decision-making.*

2.4 This problem *is even more important to solve because of the competitive landscape. Owners expect stricter deadlines, better cash flow management, and more open reporting. The World Economic Forum has consistently stated that the future of construction depends on better value-chain integration, digitization, and data-driven decision-making, rather than on project management methods that don't work together (World Economic Forum, 2016). Recent criticism emphasizes that collaboration, talent development, and robust data governance are essential to the evolution of digital construction (World Economic Forum, 2025). For contractors, optimizing inventory goes beyond just a normal warehouse issue. It becomes a strategic skill that affects project reliability, client trust, and profitability.*

2. Problem Statement

3.1 The main problem *for the business is that the contractor can't keep the demand for materials in line with the actual needs for time, cost, area, and approval status. Traditional systems rely on ERP stock balances, purchase orders, human material needs, and planner forecasts. However, these inputs are not automatically connected at the activity and area levels. As a result, site teams often only find out about problems at the start of work, procurement responds too late, and cost control gets an erroneous picture of how ready the project really is. Conventional solutions are not good enough for three reasons.*

3.2 ERP systems *initially have strong transactional capabilities, but they face operational problems. For example, they keep track of purchases, receipts, issuances, and invoices, but they don't automatically check whether the site is ready to conduct the next approved activity. SAP Materials Management has strong features for purchasing and tracking inventory, but it can't serve as a predictive scheduling engine on its own (SAP, n.d.).*

3.3 Oracle Primavera P6 *is a tool for scheduling large projects and tracking costs, and it excels at planning and managing activities. However, it does not check whether physical materials are available, whether inspection results are accurate, or whether the amounts issued versus consumed are accurate (Oracle, n.d.).*

3.4 Manual site methods *allow for engineering judgment but lead to inconsistencies in how amounts, approvals, and material requests are recorded. There are major operational problems: materials are purchased without proper alignment with actual work packs; warehouse discrepancies are not always linked to WBS, activities, and areas; quantities reported as completed may not be inspected or accepted; and rework consumes more inventory without timely oversight. There are also strategic problems.*

3.5 Management *can't tell the difference between schedule delays caused by design problems, late deliveries, low labor productivity, or unapproved work.*

3.6 Financial problems *include: overstock tying up capital, emergency purchases raising costs, unutilized labor rising because of material shortages, and the projected cost-at-completion becoming unpredictable. The paper says that the problem is not just a software bug, but a failure in information and governance.*

The organization needs an AI-based system that can turn baseline demand into work packs that can be acted on, check that materials and labor are ready, tell the difference between "quantity done" and "quantity approved," and always bring together commitments, inventory, actual problems, and site progress into one operational truth.

3. Data analytics Component

4.1 The case *has a lot of data, so it is good for advanced analytics. The main structured data sources are the contract Bill of Quantities (BOQ), the estimate build-up, the SAP material master, the buy requisitions, the purchase orders, the goods receipts, the goods issues, the commitments, the budget lines, the cost posts, and the Work Breakdown Structure (WBS) parts. Primavera gives you organized scheduling information, including activity IDs, baseline start and finish dates, expected quantities, and lookahead periods. Engineering comments, inspection notes, constraint logs, non-conformance report records, and variation order narratives are all examples of semi-structured data. Real-time or near-real-time streams include reporting quantities on tablets, scanning transactions in warehouses, and data on worker presence from RFID or IoT. Historical data encompasses prior project consumption trends, supplier lead times, rework documents, and productivity records.*

4.2 This evidence *substantiates all four analytical modalities.*

4.2.1 Descriptive analytics:

include information about current inventories, unpaid bills, delayed shipments, distributed quantities, and the status of progress.

4.2.2 Diagnostic analytics:

Look into why activities are delayed or materials are in short supply by looking at how these events are related to design restrictions, inspection failures, procurement delays, or unusual use.

4.2.3 Predictive analytics

can tell you what materials you will need in the future, when you might run out of stock, how likely it is that your supplier will be late, and what your productivity will be like.

4.2.4 Prescriptive analytics

then tells you what to buy, move, reserve, or escalate, and when to do so. The literature strongly supports the use of machine learning for forecasting demand and managing inventory.

4.3 Analytics Framework

Sukolkit et al. (2024) show that machine learning and ensemble forecasting facilitate demand prediction and inventory decision-making in the steel sector. Ayhan et al. (2024) contend that AI-driven predictions of erratic material utilization may reduce inventory costs and improve managerial decision-making. While these studies are not solely focused on construction, they are relevant given the inherent uncertainties in timing, fluctuating demand, and the need to balance service levels with holding costs in inventory management.

A practical analytics framework for this scenario would include a SQL Server data warehouse with dimensional tables for project, WBS, activity, BOQ item, area, resource, vendor, inspection status, constraint type, and version, as well as fact tables for planned quantities, progress, manhours, costs, commitments, material movements, inspections, rework, and variation orders. The main bridge would be a control matrix that links the BOQ item, WBS code, activity ID, area, productivity requirements, and material specifications. This bridge allows the system to ask management-related questions, such as: what materials are needed for the next 2-week lookahead in Zone 2? What amounts have already been allocated? What amounts are physically available and quality-approved? And what is the expected productivity if the designated crew is deployed?

4. AI-Solutions Overview

5.1 The proposed solution *is an AI-powered construction management application that focuses on optimizing inventories while also covering planning, procurement, warehouse management, quantity evaluation, and project oversight. The conceptual architecture has seven layers: tender and estimate; contract BOQ and versioning; SAP budget, commitments, and actuals; Primavera baseline and lookahead; site execution with work packs, quality assurance, RFID, and material tracking; a SQL Server data warehouse; and an AI and analytics engine that gives alerts and dashboards.*

5.2 The AI stack is mixed on purpose*.*

5.2.1 Supervised machine learning algorithms use data from past projects to guess what materials will be needed, how likely it is that supply will run out, how long it will take to get new stock, and how productivity will change.

5.2.3 Unsupervised learning sorts materials and providers into groups based on how volatile, critical, and likely they are to be delayed.

5.2.3 Semi-supervised methods are useful because construction companies often have incomplete labels. For example, only a small number of delays are clearly labeled by their causes. A rule engine improves machine learning by adding deterministic limits, such as limits on the number of items, checks on the unit of measure, required approvals, minimum and maximum thresholds, and logic that blocks work packs that are not ready for inspection.

5.2.4 Natural language processing can look at comments from inspections, site remarks, and procurement notes to sort hazards and make weekly management reports.

5.2.5 Robotic Process Automation (RPA) can make it easier to obtain approvals, send notifications, and report recurring exceptions.

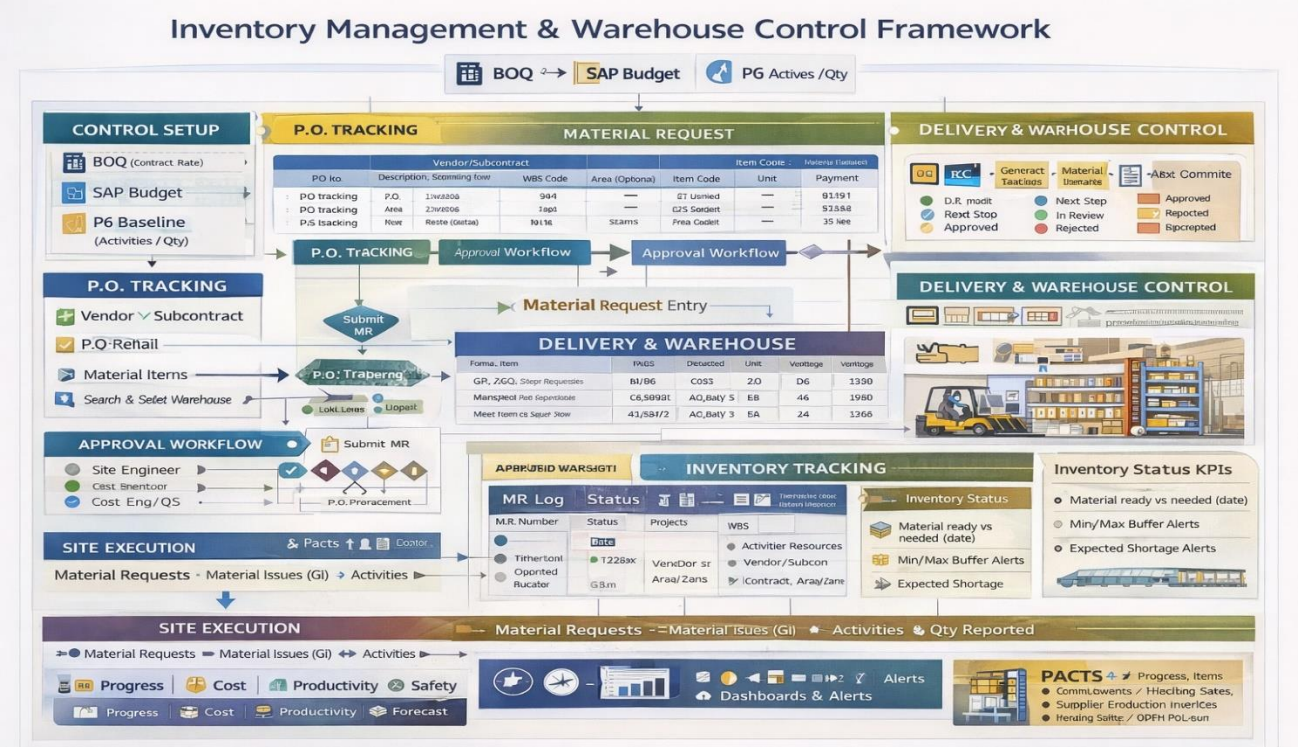
It also solves a big problem in practice: the difference between execution data and earned data. Only the correct amounts should be used to determine the value earned. Because of this, there is a measurement and quality assurance gate between site reporting and management data in the program. The flow includes the amount the site engineer did, the amount QA/QC thought was correct, the amount approved as earned, and the NCR or rework written down separately.

5.3 This governance component is very important because without it, progress would be overstated, *and the real material losses from rejected projects would be hidden. More study on artificial intelligence in construction supports this path. Adebayo (2025) notes that artificial intelligence is increasingly used in construction for decision support, predictive analytics, optimization, automation, and performance improvement. Alzubi et al. (2023) also say that automated monitoring technologies make construction more efficient by making performance factors easier to see. Collectively, these findings validate the case design: AI must not be regarded as an isolated forecasting instrument, but rather as an integral component of an operational model that interlinks sensing, transactions, and managerial actions. Conceptual Workflow of the Proposed Application: Stratum, Fundamental Procedure Standard, AI/Analytics Position, Preparation and arrangement, Bill of Quantities, estimate, SAP budget, Primavera P6 baseline, control matrix, Demand drivers delineated; standards and linkages formalized. Execution oversight, advance planning, work package creation, material requisition, and approvals.*

The rules engine verifies amounts, units of measure, readiness, and role-specific workflows. Logistics and acquisition: Purchase order tracking, delivery, goods receipt note, goods issue, and inventory by region. Forecasting shortages, lead-time vulnerabilities, and transfer/reorder advisories. Field assessment. Quantity completed, inspected, authorized, non-conformance/rework, and RFID man-hours. Detection of anomalies and monitoring of productivity trends. Management acumen. Data Warehouse, dashboards, electric vehicles, expenses, preparedness, notifications. Prediction, narrative construction, and prescriptive guidance.

5.4 Conceptual Workflow of the Proposed Application

Layer	Core Process	Typical AI/Analytics Role
Planning and setup	*BOQ, estimate, SAP budget, P6 baseline, control matrix*	*Demand drivers defined; norms and relationships standardized*
Execution control	*Lookahead planning, work pack generation, material request, approvals*	*Rules engine validates quantities, UOM, readiness, and role-based workflow*
Warehouse and procurement	*PO tracking, delivery, GRN, GI, inventory by area*	*Prediction of shortages, lead-time risk, and transfer/reorder recommendations*
Field measurement	*Qty done, inspected, approved, NCR/rework, RFID person-hours*	*Anomaly detection and productivity trend monitoring*
Management insight	*DW, dashboards, EV, cost, readiness, alerts*	*Forecast, narrative generation, and prescriptive recommendations*



6. Conceptual Analytics Techniques, KPIs, and Decision Support

***6.1 A thorough case study must** illustrate how analytics influences decision-making.*

Descriptive analytics would give you up-to-date information on things like the amount of inventory you have, how it's spread out by location, the status of your purchase orders, the expenses you've committed to, the open material requests, the approved quantities, and the amounts you've issued vs the amounts you expected. Using constraints log that sorts of delays into design, material, access, permit, inspection, personnel, or equipment, with an owner, due date, and status for each, diagnostic analytics will find the root causes of problems.

Predictive analytics would look at what materials will be needed in the future based on activity and location, how well suppliers can meet those needs, and create stockout probability distributions.

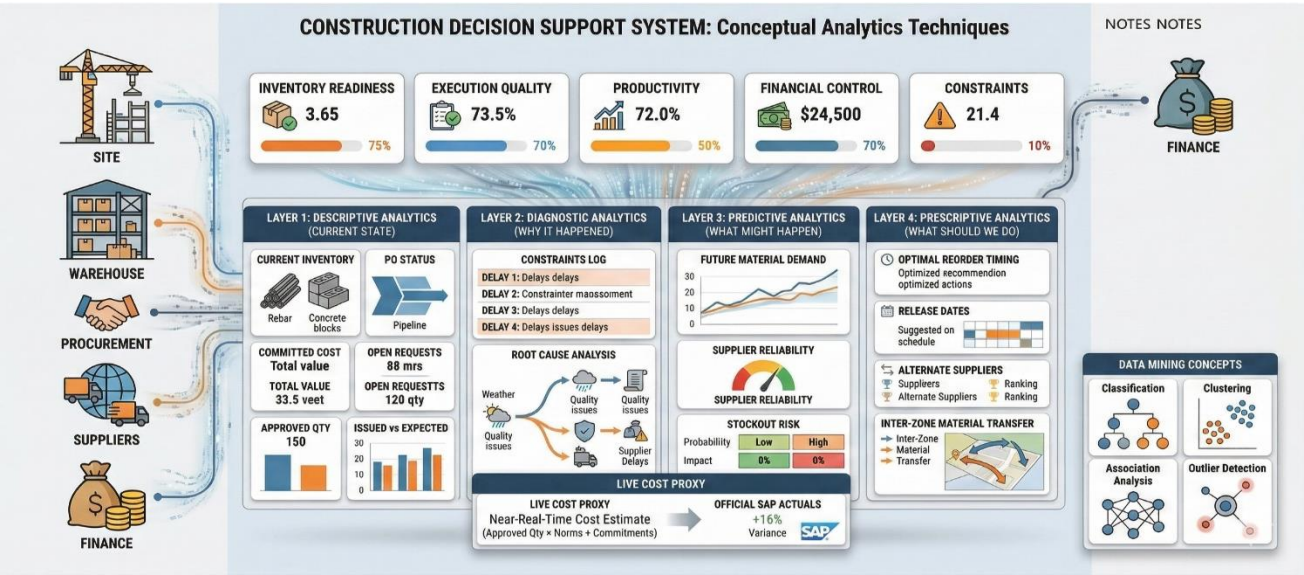
Prescriptive analytics could recommend release dates, reorder amounts, alternate sources, or moving materials between zones. The dashboard layer has to combine standard project control metrics with inventory-specific data. The material availability rate, stockout frequency, inventory turnover, holding days, on-time delivery rate, material readiness index, minimum-maximum breach count, and issued vs normative consumption are all examples of inventory KPIs. The execution KPIs include the number of finished tasks, the number of approved tasks, the number of tasks that need to be redone, the labor productivity index, and the constraint closure rate. Financial KPIs include commitments, actual costs, cost proxies, earned value, the Cost Performance Index (CPI), the Schedule Performance Index (SPI), and forecasts of what will happen when the project is done. The main idea behind the design is that inventory shouldn't just be seen as a balance of goods, but as a service that is ready for scheduled, permitted actions.

Data mining techniques make it easier to make decisions. Classification models can predict how likely it is that an activity will be delayed because of material. Clustering can sort materials into groups like critical, volatile, long-lead, or high-carry-cost. Association analysis can find common co-occurrences, such as how inspection delays often happen when certain finishing materials are reissued. Outlier detection can find strange problems with materials, waste, or questionable tendencies in how things are used. In this way, management goes from just reporting on the situation to recognizing patterns and acting.

The case may use a "live cost proxy" to make up for the known delay in ERP updates. The software should use allowed amounts, productivity standards, and existing commitments to come up with estimates of expenses that are close to real time. This is because SAP actuals usually come at the end of the month. Then, these estimates should be compared to SAP-posted actuals. This dual perspective gives managers a heads-up without replacing official financial data.

Illustrative KPI Set

KPI Group	Example Metrics	Decision Use
Inventory readiness	Ready vs needed by date; min/max breach; expected shortage alerts	Priorities procurement, reservation, or zone transfer
Execution quality	Qty done, qty inspected, qty approved, NCR/rework	Protect EV accuracy and expose hidden material loss
Productivity	Planned vs actual person-hours; productivity index; crew variance	Adjust crew composition and activity sequencing
Commercial control	Commitments, actuals, cost proxy, EV, CPI, SPI, EAC	Improve forecast and cash-flow visibility
Constraints	Open constraints by type, owner, due date, and trend	Escalate blockers before they damage the schedule



5. Analysis of Digital Transformation

6.1 The suggested system's digital transformation is important because it changes how work gets done, not just turning paper forms into digital ones. Before the change, site engineers made requests by hand, procurement responded to shortages, planners managed lookaheads on their own, and finance only got accurate cost information after delays in posting. After the transformation, the first step becomes the work pack. The engineer chooses tasks from the approved lookahead, the system figures out the expected amounts and needed materials based on standards, validation protocols check availability and role permissions, and approvals make sure the work can go ahead. This changes the process from reporting that isn't connected to being ready for coordinated execution. Automation affects many repetitive tasks. RPA can help route requests for materials, remind approvers of overdue tasks, make notifications for shortages, and make exception reports. The rules engine can stop requests that go over the allowed number of items, don't match the unit of measure, or use materials for a workpack that don't have baseline authorization. These traits make clerical work easier and improve discipline. They agree with McKinsey that digital transformation in construction should be linked to changes in methods rather than just new tools (Koeleman et al., 2019).

Organizational culture can change. When the cause of the delay is evident, such a structural problem, an inspection status, or an inventory problem, instead of an unclear story, it is easier to make decisions. Planning, warehousing, procurement, project controls, and quality assurance all come together around one operational reality, which makes cross-functional communication better. At the same time, digital revolution might cause people to resist it. People who work in the field may see more controls as more bureaucracy, while managers may worry that they will find out that their planning assumptions were wrong. As a result, the transformation strategy must include training, clear definitions of responsibilities, and clear examples of how the application reduces rework, shortens wait times, and eliminates manual reconciliation.

The customer experience in construction is indirect but real. Better material readiness and accurate progress reporting make the timetable more dependable, reduce conflicts, and build the contractor's credibility with customers and consultants.

A contractor that combines ERP, planning, and site execution data into one AI-driven control environment has a strategic edge in terms of how predictable and advanced their reporting is.

6. ERP Integration and Role

7.1 is important to the argument because it provides a financial and transactional base. In SAP, the relevant modules are Materials Management for purchasing, receiving, issuing, and managing inventory; Project System or Controlling for budgets and obligations related to WBS; and Finance for official postings. SAP's materials management system is good at making structured purchases and inventory transactions, but it needs project-specific logic to show materials at the activity and area levels, not just at the material and storage-location levels (SAP, n.d.). Primavera P6 adds the time dimension. Oracle describes P6 as a system for planning, scheduling, and managing large programs and projects. It has features for multiuser access, cost coordination, and status gathering (Oracle, n.d.). P6 gives you the basic schedule, expected amounts, work order, and lookahead periods. Being ready to carry out a plan requires more than just knowing when things will happen. The application shows a control matrix that links each relevant BOQ item to the WBS, activity, area, productivity standard, and material standard. This matrix is the link between ERP and the real-world conditions on the site. The data flows as shown. Contractual and estimating data show what the expected material coefficients and budgetary rates are. SAP keeps track of inventory balances, purchase orders, debts, receipts, distributions, and actual costs. P6 gives baseline and short-term demand forecasts. Site tablets keep track of work pack choices, daily quantities, inspection status, and material requests. Warehouse transactions confirm receipt, reservation, transfer, and issuance. The data is put into a SQL Server warehouse, where dimensions and facts are made the same for each project, date, WBS, activity, BOQ, area, vendor, and version. After that, the AI layer looks at the warehouse and sends forecasts, alarms, and suggestions to dashboards or workflow queues. ERP integration needs formal control techniques. Because versioning is a part of every project, the system needs to keep track of changes to the budget, variations, the baseline, and the time impact evaluations. Audit trails must show who made modifications, what those changes were, when they were made, and why they were made. Cut-off criteria are needed to make sure that daily or weekly reports are closed and reconciled on time. Identity and permissions are very important. Engineers, QS staff, QA/QC workers, warehouse controllers, planners, and cost controllers should only be able to access and authorize things that are part of their jobs. Role-based access control is a well-known idea in information systems research, and it's relevant here since project data includes financial, operational, and human domains (Ferraiolo and Kuhn, 1995; NIST, 2016).

7. Business Impact & Value Creation

8.1 The expected company value can be broken down into three types of results: financial, operational, and strategic. The system can help with emergency purchases, extra inventory, wasted materials, and inefficient work caused by supply shortages. Optimized procurement scheduling improves cash flow efficiency since the contractor gets resources closer to when they are needed, while keeping service levels up with predictive safety stock. McKinsey says that when operational models are changed to put data and execution control first, digital transformation in construction can lead to big gains in productivity and savings on costs (McKinsey, 2019). Each contractor must check their own numbers, but the trend of advantage is quite strong. The proposed system makes operations more dependable. Activities begin when everyone is ready, not when they think they are. Instead of getting materials from a general inventory, they are requested through work packs that are linked to specific scheduling needs. The Work Breakdown Structure (WBS), activity, and location are all connected to warehouse transactions. Rework is clearly documented rather being hidden in overall consumption. Instead of looking back at past performance, workforce productivity is measured against set baseline norms. These changes reduce delays, improve collaboration, and strengthen short-term management. Instead of a solution for a single project, the system sets up a scalable digital platform. Once the organization has set up the control matrix, data model, and governance rules, it can compare suppliers across projects, measure productivity by trade and location, and add up the inventory risk across a portfolio. In the medium term, the platform can help with digital twins, BIM-integrated quantity validation, and generative AI assistants for questions about project control. Scalability is important because it increases the value of AI when learning and governance are shared across projects instead of starting from nothing with each new one.

8. Risks, Ethical Issues & Limitations

9.1 its potential, the proposed system raises fundamental questions of governance and ethics. The most important thing is the quality of the data. If WBS codes, activity IDs, material masters, units of measure, or area structures are wrong, the AI layer will make the problems worse instead of better. Second, there is a lot of model risks. Predictions based on bad or missing historical data may fit too well with past situations and fail when changes are made to a project's design rationale, site logistics, or procurement market. There is a lot of danger involved in adopting. Construction workers may bypass the system if they find it too complicated or if it makes it hard to make important decisions on the site. Privacy and security are equally important. RFID and execution data related to the workforce could make it easier to see how productive people are, but they could also present privacy issues if too much data is collected or used without clear purpose limits. So, we need information security measures. ISO/IEC 27001:2022 outlines a risk-based framework for managing information security.

This is relevant to this case because the proposed application handles data about expenses, procurement, suppliers, sites, and staff (ISO, 2022). Access management, secure development, logging and monitoring, supplier security, backup and recovery, and change management are all examples of essential controls. Role-based permissions must restrict data visibility to legitimate business needs, especially for workforce-related documents. Bias is another issue that needs to be addressed. If procurement or performance models are developed using historical data that shows old supplier preferences or inconsistent inspection techniques, they may keep wasteful patterns going. Managers should see AI outcomes as helpful tools for making decisions, not as facts that can't be wrong. Human evaluation is necessary, especially for changes in the effects of orders, claims made under contracts, and unusual events that happen during a project.

9. Recommendations & Future Scope

10.1The scenario is set up as an integrated conceptual model, but putting it into action requires phased implementation, data cleaning, user training, and possibly custom SAP and mobile integration work. The business case is strong, but benefits can only be realized via discipline in execution.

Suggestions and Possible Directions:

The first suggestion is to carry out the solution in steps. The contractor should start a trial project that only focuses on important materials like rebar, blockwork, gypsum, or long-lead MEP items. The pilot must include the control matrix, work pack-driven material requests, QA approval checkpoints, commitment transparency, and a limited dashboard set that focuses on readiness, shortages, and issued versus normal consumption.

This strategy reduces change fatigue and quickly shows benefit. Before the organization can grow, it needs to standardize its master data. This means making sure that the WBS codes, activity names, hierarchical structure, units of measurement, material master references, vendor IDs, and version control methods are all in sync. Master-data governance should be seen as a strategic initiative on its own because construction data failures happen more often because of inconsistent keys than because of bad models. Before the contractor can start working on advanced AI projects, they need to set up the data warehouse and governance layer. This includes the project's dimensions, the Work Breakdown Structure (WBS), activities, the Bill of Quantities (BOQ), the area, the version, the inspection status, the type of constraint, the vendor, and the crew. It also includes data about commitments, material movements, inspections, variation orders, rework, projected quantities, progress, man-hours, and costs. Once strong foundations are in place, management may feel more confident about using machine learning models for stockout risk, demand forecasting, and supplier performance. Fourth, management must see the measurement and quality assurance gate as very important. The permitted quantity, not just the reported number, should be used to figure out the value earned. You need to keep an eye on NCR and rework separately since hidden rework makes productivity and inventory estimates less accurate. Before releasing SAP month-end actuals, the business should set up a real-time cost-proxy layer to help with operational decision-making. Sixth, teaching and managing change must happen at the same time as technological implementation. Users need to understand how to use the system and why the new controls are in place to protect schedule reliability and cost transparency.

The future is really bright. You may add BIM-integrated material and location data, computer vision for yard inventory and progress verification, digital twins for scenario analysis, supplier evaluation models, and portfolio-wide demand projections to the platform. You can make NLP better by using it to look into comments from consultants, site logs, and claim conversations. Eventually, the contractor may include a generative AI assistant in the warehouse and controls layer. This would let managers ask simple questions like "Which operations in Zone 3 are at the most risk next week?" or "Which suppliers are most likely to cause delays?" The integrated framework established in this case study would be the basis for such extensions.

Final Assessment: This case study says that AI-optimized inventory management in construction is mostly a problem of integrating time, cost, quantity, quality, location, and approval status. The proposed solution does not see AI as a standalone forecasting tool. It uses machine learning, rule-based validation, robotic process automation, and advanced analytics in an operational architecture that works with both ERP and schedules. The contractor may go from reactive inventory management to predictive execution preparedness by combining SAP transactions, Primavera baselines, site work packs, QA gates, RFID-enabled monitoring, and a SQL Server data warehouse. The main point of the example is that the problem with many construction systems is not that they don't have enough data, but that they need better connectivity, governance, and scheduling. The proposed platform might make schedules more reliable, cost predictions more accurate, warehouses more efficient, and management more trustworthy if it is rolled out in stages with strong master data, security safeguards, and user involvement.

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